

Towards Social Video Verification of Face Expression, Identity, and Lighting for DeepFake Detection

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Abstract

We explore methods to detect manipulated media of people speaking—often called *deep-fakes*—using information from a set of cameras simultaneously recording the event. Beyond existing deep learned detection methods trained on known fakes, so-called ‘social video verification’ would add another tool to help verify truth. We present a method to compare facial features from a multi-view camera setup. This uses a graph-based detection mechanism that improves performance over an existing hierarchical clustering baseline method. We experiment using expression, identity, and lighting features across two datasets, and demonstrate the efficacy of exploiting video information in a social setting. Finally, we discuss the limitations of these current approaches to this setting.

1 Introduction

Recent techniques in video generation and manipulation have made it feasible to render partially or fully faked videos of persons speaking that are perceptually indistinguishable from real footage. These fake videos—or deepfakes—have potential to cause harm. An adversary can create a deepfake which supplants original video content with controversial political views, leading to potential legal and social repercussions for the video’s victim, as in the 2015 Varoufakis “middle finger” incident [1, 26]. If reliable video verification tools are not established, distrust may grow in real videos with political, social, and material consequences [3].

The development of multi-pronged approaches for media verification is paramount. One such approach aims to limit how manipulated media can propagate online by establishing fingerprinting standards. These media provenance tools help ensure trust for established platforms, distributors, and creators (e.g., the Starling Project or C2PA Initiative [8]). Another such approach aims to automatically detect fake media. These detectors are in an ‘arms race’ with forgery generation methods. Generalized deep learning based detectors are not currently robust; their performance has been shown to decay when the generation method is unknown (e.g., $2.7\text{--}6\times$ less [15]) due to overfitting to their training data. Falsification methods will increasingly avoid gross artifacts both at a low level, such as created by noise, blur, compression, or misaligned edits, or at a high level, such as mismatched jewellery, eye reflections [24], or erroneous facial blood flow [9, 30]. As such, Tolosana et al. [25] describe a need for improved detection methods, but also for work in providing new tools for different settings that might exploit additional recordings available at capture time or afterwards.

Towards these efforts, we investigate the social video verification setting [24] that attempts to verify events through the social consensus of recording devices, and so build a potentially useful tool

for a class of videos of public events. The high-level aim of this system is to prevent the circulation of fake videos under the pretense that they actually occurred by using other views of the event as might be captured by a crowd of people. While audio is an important cue here [9, 13], we focus on video, e.g., for cases where the intention or meaning of a speech is changed by visual manipulation.

Given video of a person speaking, we wish to identify which videos, if any, are being faked and at which times. We check videos for consistencies in expression, identity, and lighting by extracting features for detection (facial landmarks, latent identity embeddings, and spherical harmonic lighting coefficients), then use temporal aggregation and clustering to separate the cameras into real and fake sets. For evaluation, we generate fake examples using state-of-the-art manipulations from the recent literature. We generate fake examples by using viseme generation for different audio [63], face replacement [22], and portrait relighting [45]. Then, we indicate the detection accuracy by our proposed consistency metrics, showing potential for this kind of approach as a future analysis tool.

2 Related Work

Given its importance, there are many recent works on deep fake detection. We refer the reader to wider surveys of the field [17, 40, 43], and here select a subset of works.

Deepfake creation Current manipulations can synthesize a fully fake face [18], swap identities [22, 40], and manipulate specific facial expressions [20]. Many methods focus uniquely on identity swapping, or on a mix of identity and expression swapping [29]. Expression transfer exemplifies more subtle manipulation than identity swapping, and can be accomplished with auto-encoders [51], or with GANs paired with a variety of problem-specific losses and modifications [83, 86, 44]. Renukanand et al. use deep networks to generate a talking face given a single image or video and desired audio, e.g., to automate language-agnostic translation with lip synchronization [83, 84].

Some works do not use deep learning. Thies et al. create Face2Face to transfer the expression of a source actor to a target video in real-time [89]. This method is used to generate some of the fakes in the FaceForensics++ [65] dataset. Averbuch-Elor et al. warp portrait photos to mimic a source video, and transfer fine details like wrinkles to heighten realism. This work can generate convincing manipulations as long as there are no large changes in head pose [6].

Finally, Tolosana et al. [40] define ‘Gen 1’ category automatic fakes as those that still produce visible artifacts, and so far almost all approaches fall into this category [68]. This is especially so as the complexity of the underlying face appearance increases, e.g., strong profile view, highly dynamic expression. Arguably the closest automatic approach so far to ‘Gen 2’—without visible artifacts—is the FaceShifter model of Li et al. [22],

Deepfake detection Detection methods primarily focus on finding artifacts created by sensor noise, compression, or edits. These often use neural networks to decide whether or not a single video has been modified, which can overfit to their training data—the performance of many detection methods decays for in-the-wild scenarios that have larger variation in compression, noise, and blur [40, 44]. For instance, the work of Guera et al. combines a CNN for feature detection and an LSTM for temporal information to predict if a video is fake given two seconds of input [10]. Li et al. look for face warping artifacts caused by resolution changes between source videos and fake output videos [25]. Neves et al. show that GANs leave ‘fingerprints’ in the high frequency information of their output videos. They show that many existing detection methods rely on detecting these fingerprints, and demonstrate that detection accuracy drops dramatically when they remove them [28].

Detectors can also be trained on subject behaviors. Li et al. look at eye blink rates and abnormal physiological cues [24]. Agarwal et al. show that fakes can be detected by looking for consistent action unit models for persons of interest. They codify a speaker’s normal behaviors as

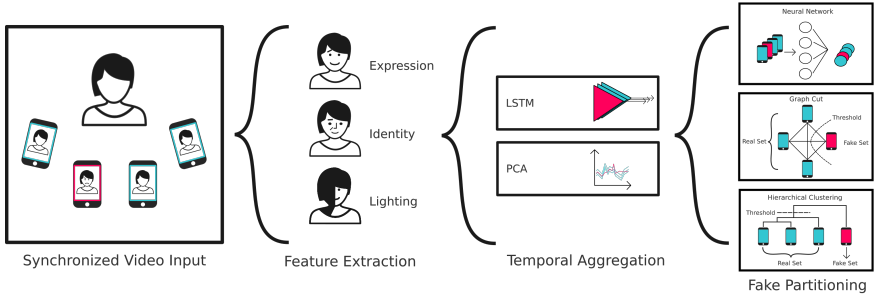


Figure 1: Overview of our scenario and evaluation setup. A set of synchronized video captures contains fakes across expression, identity, or lighting. We compare metrics with different partitioning approaches to determine which cameras belong to the real and fake sets.

high-dimensional clusters, and detect fakes if they fall outside these clusters [9]. Such approaches require an existing corpus of data for a particular speaker. Agarwal et al. match phoneme and viseme pairs to detect expression or audio forgeries [10], and Hosler et al. consider mismatched video and audio emotion [11].

Detection can be aided by creating and publicly releasing data. Khodabakhsh et al. create a dataset of manipulated images—Fake Faces in the Wild [12]. Rossler et al. create a dataset of fake videos called FaceForensics++ [13], and determine that expression transfer methods like Face2Face are most difficult for humans to detect—this motivates the need to detect subtle manipulations in image content. These datasets are all single view, so are insufficient for the social verification problem. Both Tursman et al. [14] and Jiang et al. [16] introduce datasets captured with multiple cameras with sets of diverse actors in a close-up shot setting; we use these to evaluate detection methods.

One final consideration for detection methods is interpretability. Verdoliva [15] describes that neural network detection methods are hard to interpret, which is insufficient for scenarios requiring the interpretation of detected manipulated video, e.g., in legal contexts.

3 Method

Social video verification assumes a set of C capturing cameras $c_1, \dots, c_i, \dots, c_C$ viewing the same event in real time, with a subset of real videos R and faked videos F such that $|R| > |F|$ (Fig. 1). The task is to identify and separate the set of real videos from fake videos.

We assume that frame-accurate synchronization is provided by communication between cameras via peer-to-peer wireless networks [9]. Given the setting, it is important for any detection method to operate efficiently in compute and communication size, i.e., to not require the transfer of video frames between devices, such that C can be large. Our goal is to extract a compact set of features *per camera* that can then be used to efficiently cluster cameras, such that video frames do not have to be shared between cameras. In Sec. 4, we will describe experiments with specific features for facial expression, identity, and lighting.

Given a set of features per frame $\{\vec{m}_{c_i,1}, \vec{m}_{c_i,2}, \dots, \vec{m}_{c_i,n}\}$, where each camera c_i captures a video of n frames, we consider how to integrate temporal information to capture variance differences between views via functions over windows in time (Section 3.1). Then, we describe methods to separate our cameras into real and fake subsets (Section 3.2).

3.1 Describing Features over Time

Oftentimes, clues over time can reveal a fake. Sometimes, features extracted from images may show temporal instability. For both of these reasons, we describe methods for time.

For each camera c_i , we define an $f \times n$ matrix M_{c_i} that stores the set of features with f components each extracted over a window size of n frames:

$$M_{c_i} = \begin{bmatrix} \vec{m}_{c_i,1} \\ \vec{m}_{c_i,2} \\ \vdots \\ \vec{m}_{c_i,n} \end{bmatrix} = \begin{bmatrix} m_{1,1} & m_{2,1} & \dots & m_{f,1} \\ m_{1,2} & m_{2,2} & \dots & m_{f,2} \\ \vdots & \vdots & \ddots & \vdots \\ m_{1,n} & m_{2,n} & \dots & m_{f,n} \end{bmatrix}.$$

Classical PCA To represent variation compactly as a signal over time per camera c_i , one approach proposed by Tursman et al. [12] is to perform classical PCA on each of the matrices $M_{c_1}, \dots, M_{c_C}$. At each frame, we represent the frame as the distance from a feature's projected position in the PCA subspace for M_{c_i} to the center of that same PCA subspace. We calculate this length using the Mahalanobis distance metric, which can be used to find outliers in multivariate comparisons. We define this $n \times 1$ distance vector for c_i as $\vec{d}_i$, where:

$$\vec{d}_i = \sqrt{\sum_{cols} (M_{c_i}^{proj} - A)^T cov(M_{c_i})^{-1} (M_{c_i}^{proj} - A)} \quad (1)$$

$$M_{c_i}^{proj} = (M_{c_i} - \bar{M}_{c_i})E \quad (2)$$

Intuitively, quantifying and comparing feature variance over n frames intends to ignore absolute appearance factors. For example, to detect unusual face motion, this approach ignores the pose of the head but still captures whether face parts are moving in similar ways.

Given the occasional noise in feature representations, we also compare to a robust PCA approach [13] (with details in supplemental material). This is significantly more expensive to compute but can accommodate outliers within M .

3.2 Separating Real and Fake Camera Subsets

Given a set of distances D , we wish to separate the real and fake subsets. We assume the scenario where fakes in F are manipulated in a consistent way, and so should group similarly.

Hierarchical clustering We include a baseline approach followed by Tursman et al. [12]. Given a set of vectors $\vec{d}_1, \vec{d}_2, \dots, \vec{d}_C$, we obtain a weighted binary tree t by single-linkage hierarchical clustering, using the L^2 distance between pairs of tree leaves to repeatedly join the minimal distance clusters. Motivated by the idea that real and fake camera clusters should be far from each other, we consider a variant that computes the sum L^2 feature distance to *all other* leaf nodes.

$$d_i = \sum_{c_j \neq c_i} \sqrt{\sum_{k=0}^n (\vec{d}_{c_j,k} - \vec{d}_{c_i,k})^2} \quad (3)$$

Given a clustering, we use the final tree's shape and costs to determine which, if any, of the inputs are faked by inconsistency with the majority (Fig. 1). For a tree t with k connections and a distance weight w per connection, we calculate:

$$\phi = w_k / w_{k-1}, \quad (4)$$

where ϕ represents the ratio of the cost of the ultimate and penultimate connections in t . If ϕ is below this threshold, then we say that there are no fakes present. If ϕ is greater than a set threshold τ , we

say that there is inconsistency within our input set of cameras, and at least one fake is present. To find which cluster is the real cluster, we re-cluster the tree into two sets and pick the cluster with the smallest overall distances. We set τ between 1.05–1.9 empirically based on labeled datasets and features.

Graph cut with distribution-reweighted edge distances. We also compare to an approach based on a graph weighted by the probability that real and fake cases will lie in different modes. For each timestep, we model each camera as a node in a fully connected graph, to be cut into the two real and fake subgraphs. We expect edge distances to have separate distributions—larger distances are likely to result from two nodes belonging to different real/fake subgraphs. We analyze the distribution of all feature distances. If the distribution is bimodal, then we anticipate a fake; conversely, if the distribution is unimodal, then we do not.

We use a dip test [12] to find the probability that our data is unimodal, and use it to split our set of distances into subset distributions. Since the within-class-subgraph distribution, L , will have smaller distances, it will be the leftmost distribution discovered by the dip test. Then, we find the probability that each edge distance d could have taken on its value or greater assuming it belongs to L :

$$p = P(L > d) \quad (5)$$

We want edges in our graph to have a high weight when p is low—edges that connect real and fake nodes should have the largest weight. As such, each edge weight E is set according to:

$$E = \frac{1}{p^2} \quad (6)$$

Finally, given the distribution-reweighted edge distances, we find the maximum cut in the graph to partition the nodes into a real and fake set.

To handle isolating the zero fake case, we use the distributions analyzed in previous steps. First, across all possible cuts, we consider the likelihood of the max cut occurring, p_m . Then, we combine it with the probability of the feature-distance distribution being unimodal, p_u . Our time instance is then given a value $T = p_m \times p_u$. If $p_m p_u$ is above a threshold τ , then we consider the set of videos to all be real. As before, we set τ between 0.01 and 0.10 empirically.

3.3 End-to-end Learned Separation

Finally, we describe a baseline learning approach that separates cameras into real and fake subsets. This method uses *generic* visual features. First, we run a ResNet18 model pretrained on ImageNet upon each image and extract features from the last layer of the network. These features are flattened into a 32,768-dim vector and fed into an LSTM network with 1,024 hidden units to aggregate information over time. The hidden state of the LSTM is then taken as a summary vector at each timestep and fed into a layer of 1000 fully-connected neurons. These output to a set of logits, one per camera, that output a binary classification of real or fake. Within this model, the analogy to the other methods is that the LSTM acts as a temporal aggregator (Sec. 3.1), and the fully-connected layer acts to separate the real and fake subsets (Sec. 3.2), except that both are trained simultaneously.

4 Experiments

Our experiments consider falsifications that alter identity (Sec. 4.1), expression (Sec. 4.2), and lighting (Sec. 4.3). The classic ‘face swap’ setting of deep fakes is the same as our identity setting. The expression and lighting experiments show more subtle manipulation while preserving identity.

Tursman et al. [12] dataset The dataset consists of multi-view videos of 24 human speakers set against a black background curtain. Six cameras are set up in an arc of 65° facing a seated participant. To induce natural head and facial movement, the subject is prompted to answer random questions while being video captured. Chance performance to guess correctly the real or fake nature of each of the videos is about 1.6%.

DeeperForensics [16] dataset The dataset consists of multi-view videos of 100 human speakers in front of a dark green curtain. Seven cameras are set up in an arc of approximately 120° facing a seated participant. Subjects are requested to speak naturally with 8 target expressions. Chance performance to guess the correct real/fake videos is about 0.8%.

4.1 Identity Transfer Detection

Common ‘face swap’ approaches change the identity of a person but maintain expression and lighting, with the goal of producing a highly-realistic human puppet.

Fake generation We use the FaceShifter approach of Li et al. [12], which has no public model or official implementation presumably as a cautionary measure. As such, we retrain a model from scratch using an unofficial architecture implementation [12]. We train this on the combined CelebA-HQ and FFHQ datasets (100k images) on two NVIDIA Quadro RTX 6000 GPUs for a total of 200k iterations, 8 epochs. Then, we use this model to falsify each video view of the recordings of 15 different participants from the DeeperForensics dataset.

Fakes were generated for entire video sequences. At evaluation time, we splice together the real and fake videos, swapping out the middle third of each fake video for its respective real camera as before. Using windows scanning over the videos, we are able to test how robust our detection method is to different portions of the windows being fake.

Feature extraction To find features that would reveal identity transfer between cameras, we turn to methods designed to identify individuals within the problem of face recognition. The goal

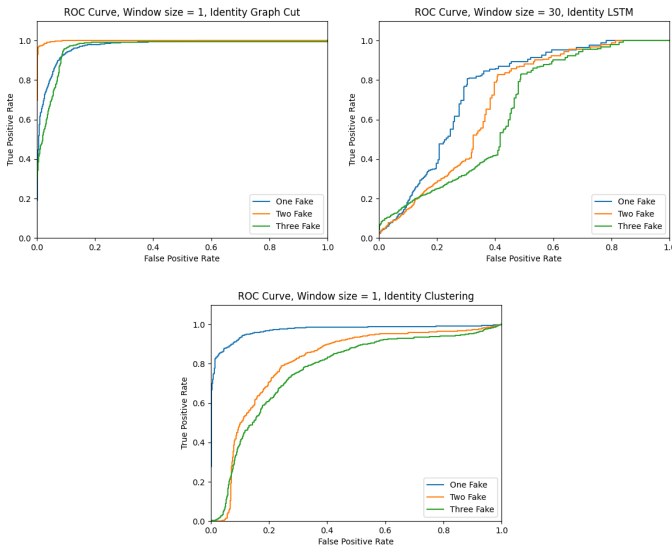


Figure 2: Performance on the identity dataset. Left: ROC curves of each method for zero to three fakes. Right: Accuracy numbers of each method for each number of fakes.

is to find a feature that encapsulates the subjects’ identities while also being invariant to changes in pose, lighting, and expression. We use a CNN-based method from Deng et al.¹ [10] to extract 512-dim. identity embedding for each frame in each camera.

Results Figure 2 presents our findings. The graph cut method significantly outperforms the other methods in all but the 1 fake case, where it performs similarly to hierarchical clustering, as a significant imbalance in the number of real and fake cameras hurts graph cut performance. The LSTM is able to achieve reasonable results because differentiating between identities is more akin to the classification tasks ResNet was trained on; however, it still lags behind the other methods.

4.2 Expression Edit Detection

Expression editing allows subtle manipulations of a face such as changing the mouth shape, raising an eyebrow, or adding or removing blinks [20].

Fake generation To detect expression edits, we consider the falsification scenario of mouth visemes that have been altered. To generate our data, we use the LipGAN method of Prajwal et al. [15] that takes arbitrary audio and a face video and generates the corresponding edited face with appropriate visemes. For each dataset, we splice in falsified subsequences generated with audio from nursery rhymes. Each video is split into three, with the middle third swapped. Rather than aiming to classify whether a whole video is fake, this simulates a scenario where any time instance in any camera may be fake or real, and our job is to declare at each time instance whether a fake is present.

Feature extraction First, we detect a face region and then fit landmarks per video per frame using 2D FAN [4]. This produces 68 2D points (x,y) on jaw, mouth, nose, eyes, and eyebrow regions. As such, we can isolate and normalize our landmarks independently by face part for our

¹<https://github.com/deepinsight/insightface>

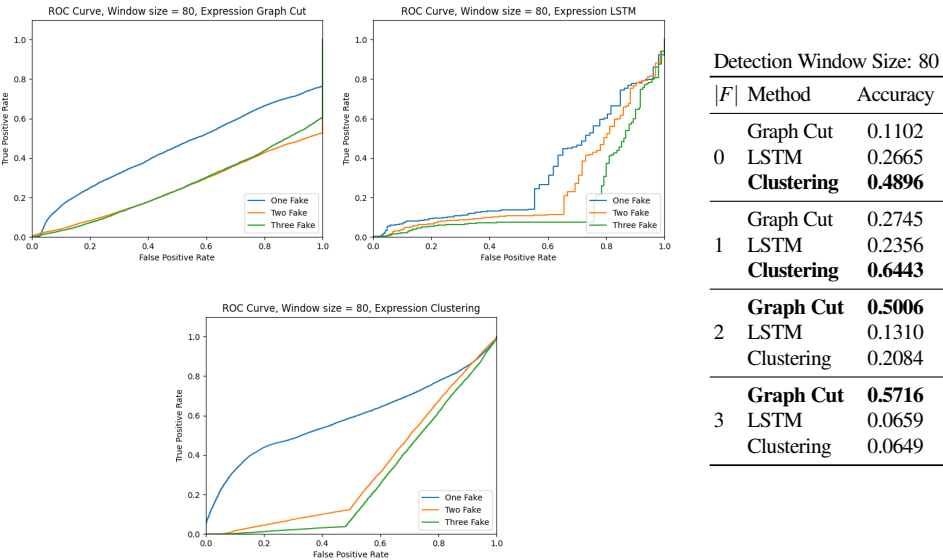


Figure 3: Performance on the expression dataset. Left: ROC curves of each method for zero to three fakes at window size 80. Right: Accuracy numbers of each method for each number of fakes at window size 80.

metric determination and clustering steps; for visemes, we consider the mouth part to detect fake motion generated for the swapped-in speech. Given the 2D points, we form M such that each $m_{i,j}$ has (x,y) pixel coordinates per frame.

Alternatives. While it might be instinctual to consider 3D feature points, often current automatic 3D fitting processes for models like 3DMM or FLAME [23, 57] can be more inaccurate than 2D landmarks and fail to track motion as well due to constraints of the underlying shape model [42].

Results Figure 3 shows that the graph cut is significantly better at partitioning two roughly equally-sized sets, but struggles when one set is much larger than the other. As the sizes of the sets become more unequal, the weight of each edge contributing to the max cut must be that much more of an outlier. The performance for the graph cut method when there are 0 fakes comes down to thresholding; increasing 0-fake performance will come at the expense of performance with more fakes. The LSTM is unable to match the the performance of the other two methods.

4.3 Relighting Detection

Modern deep learning has enabled face relighting for enhancing portraiture [21], which also allows for malicious falsification. This might seem less pernicious than modifying expression or identity, but modifying face lighting can alter the mood and so implied intent of a scene, or in principle alter the location that someone appears to be in with additional background changes.

Fake generation We implement the Deep Single Image Portrait Relighting method of Zhou et al. [45] to create our relit faces, and apply it to video frame by frame. This method can change large-scale lighting such as the major incoming angle of light, but cannot induce fine-detail or high-frequency effects like shadows from external objects. We fake the three cameras that are closest to frontal. We generate a dataset based on an ‘opposite’ lighting condition that harshly relights the face from the opposite direction as the estimated incoming direction, creating a large discrepancy in lighting.

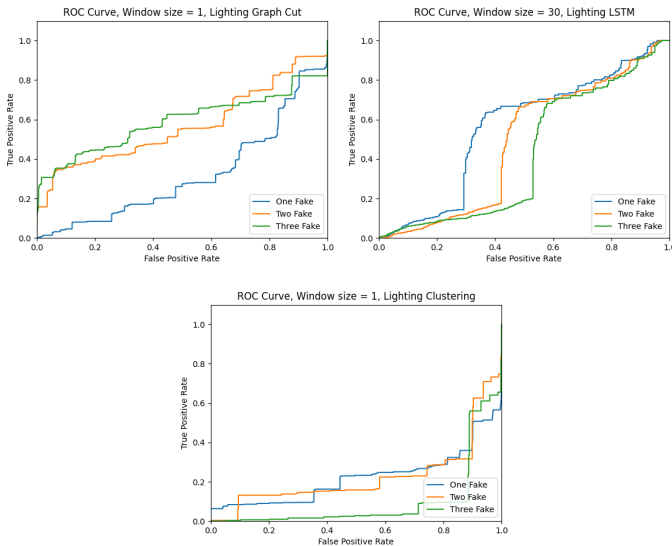


Figure 4: Performance on the relighting dataset. *Left*: ROC curves of each method for zero to three fakes. *Right*: Accuracy numbers of each method for each number of fakes.

Feature extraction We represent light incoming to a face as 2nd-order spherical harmonics (SH), estimated by treating each face as a light probe for the environment. The 0th-order coefficient represents the overall intensity of the lighting, and the 1st- and 2nd-order coefficients represent frequency variations across the sphere. We again use the method of Zhou et al. [45] for feature extraction as its intermediate representation is SH coordinates.

In many scenes, the lighting environment typically does not change through the timescales of a short video, e.g., most indoor environment captures of a person giving a short speech. In this case, analyzing temporal variation (via PCA) is not expected to help distinguish fakes.

Finally, as this model transforms each frame independently, the videos that it generates flicker slightly from frame to frame due to inconsistent lighting estimation; this is a limitation of current relighting methods. Given that the lighting environment typically does not change, in the L^2 metric case we remove this jitter in detection by computing the mean SH coefficients over the window.

Results Fig. 4 shows the graph cut method significantly outperforms the other methods across all number of fakes. As in the other experiments, graph cut performs worst in the one fake case. The LSTM again performs poorly, which is expected as our frozen ResNet was not trained to be able to distinguish between lighting conditions.

5 Discussion

The social verification setting poses new problems of finding reliable features and metrics that can identify changes in videos across cameras. In general, precisely and repeatedly estimating the real or fake nature of each camera is challenging. Over naive methods and baseline approaches [42], the graph cut method show promise.

Limitations Concerning fake generation, the performance of current state-of-the-art methods in all three experiment cases are still some ways off from not being perceptually noticeable by a human. This can be exacerbated by the multi-view case as faces turn to be more in profile. For instance, FaceShifter must see both eyes of the participant or else it will fail, and overall quality generally improves as the face becomes more frontal. Further improvement to generated fakes would help remove any performance gained from anomalies in the data.

Concerning specific features, for expression landmarks we use only 2D features as 3D landmark estimation, which can be unreliable or inaccurate. Future work to align accurate 3D estimates may improve performance. Similarly, our current SH model assumes that the lighting is centered around the face, and explicit pose modeling could align these SH coefficients. While the expression and identity features are robust to moderately wide field of view across all cameras, for lighting we expect to suffer performance decay as the baseline increases within the scenario because the feature extractor assumes the face to be relatively frontal in the frame.

Finally, each experiment operates independently, and any system to be deployed in the real world would be expected to detect any of expression, identity, or lighting variation in combination.

Conclusion Social video verification is a potentially useful setting for detecting faked images from crowd-sourced video. Toward this goal, we have investigated whether state of the art expression, identity, and lighting edits made to videos of speakers could be detected using state of the art features. Across three experiments, we find that a graph cut measure with distribution-reweighted edge distances shows promise to separate the sets.

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